

GK Bytes



Name

Class Section



8

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(M.Tech)

FriendMate Books

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Preface



GK Bytes is a series of general knowledge books that are fountain head of knowledge for the children of class 1 to 8.

This series is as much about asking the right questions as it is finding out the right answer. It is about going beyond a few select subjects, and becoming more aware of the world around us, nothing things that are new, and changing, sharpening and feeding our curiosity, and learning to get to the bottom of things. Only a questioning mind can become a truly knowledgeable one.

Some of the features of this series are :

- New contents at each level are the extension of the class curriculum, with varying degrees of challenge, interest and colour.:
- Smooth and gently progressive gradation of contents.:
- New activities to encourage individual and group tasks.:
- Enhance language skills.

All in all, the new edition ensures that learning comes about at the initiative of the learner. It does this in a way that interactively and excitingly imparts information and knowledge which is closely related to the interests, the experience, the life and world of the child.

— Author and Publishers



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CHAPTER

World of Lakes

Read the clues and find the names of the lakes in the word grid.

R	T	G	M	S	N	D	I	R	L	A	K	E	C	L	E	O
N	C	E	Q	X	H	L	O	N	A	R	L	A	K	E	Q	K
K	O	M	A	X	K	M	N	R	L	L	A	K	E	B	D	Y
J	H	U	S	S	A	I	N	S	A	G	A	R	L	A	K	E
L	T	X	J	M	E	G	R	E	A	T	L	A	K	E	S	Z
A	S	J	G	T	L	R	S	F	W	I	G	H	Z	O	Z	A
K	O	X	L	A	K	E	V	I	C	T	O	R	I	A	U	C
E	M	M	A	N	S	A	R	O	V	A	R	L	A	K	E	A
T	O	T	T	G	Z	T	M	C	K	U	E	N	I	Y	F	S
I	R	T	W	Y	S	S	Z	X	G	M	H	T	S	E	S	P
T	I	S	D	N	X	A	J	I	R	G	S	N	Z	Z	H	I
I	R	G	R	K	M	L	A	K	E	V	O	S	T	O	K	A
C	I	T	H	S	Q	T	R	T	Q	N	B	Q	B	K	P	N
A	M	D	O	D	E	L	Z	B	D	A	L	L	A	K	E	S
C	C	H	I	L	I	K	A	L	A	K	E	R	R	I	C	E
A	L	O	K	T	A	K	L	A	K	E	J	Y	X	D	K	A
H	T	V	L	A	K	E	B	A	I	K	A	L	J	X	S	E

1. This lake in Srinagar, adorned with sailing shikaras, is nicknamed 'the Jewel in the Crown of Kashmir.'
2. Located in Odisha, this is the largest saltwater lake in India, and believed to be one of the largest coastal lagoons in the world.
3. This lake in Tibet, with the Mt. Kailash standing tall in the north, is regarded as one of the holiest lakes in Hinduism as well as Buddhism.
4. This lake in Andhra Pradesh was built by Hazrat Hussain Shah Wali in 1562. In the middle of the lake is a large monolithic statue of the Buddha, built in 1991.

5. A saltwater lake in Maharashtra, this was a crater which was created by meteor that hit the Earth millions of years ago. 
6. This lake, located in Manipur, is the largest freshwater lake in North-East India, and is also known as a floating lake due to the floating phumdis or floating mass of vegetation on it. 
7. Located in Siberia, this is the deepest and oldest freshwater lake in the world. It contains about one-fifth of the world's supply of fresh water.




Helpbox



Lake Baikal, Lonar Lake, Dal Lake, Hussain Sagar Lake, Loktak Lake, Chilika Lake, Mansarovar Lake



The Dead Sea is a very salty lake that lies in a sunken rift in Israel and Jordan. With a surface of 400m below sea level, it is the lowest water body in the world.



Activity

Find a lake near your city. Find out how the lake was formed. Discuss your finding in class.

2

CHAPTER

Mammals Mania

A mammal is an animal that breathes Air, has a backbone and grows hair at some point during its life. In addition, all female mammals have glands that can produce milk. Mammals are among the most intelligent of all living creatures.

Read the clues given, find the answers hidden in the word grid, and encircle them.

H	I	B	E	R	N	A	T	I	O	N	O	R
M	F	S	K	U	L	L	N	U	L	M	N	E
A	I	M	L	L	S	A	W	Q	B	K	A	D
B	K	L	X	G	W	A	L	R	U	S	L	K
I	O	R	K	D	T	Y	Q	D	G	O	W	A
R	A	S	C	A	V	E	N	G	E	R	S	N
D	P	W	I	A	O	L	R	R	F	X	L	G
S	I	U	F	P	O	T	T	E	R	O	Q	A
R	U	O	T	C	S	G	F	S	T	Y	S	R
M	A	R	S	U	P	I	A	L	S	W	L	O
T	Y	X	A	B	Q	N	L	N	P	G	P	O



QUIZ TIME

1. What is the bony case protecting a mammal's brain called?

2. What usually is the first food of a baby mammal?

3. What is the world's largest marsupial?



4. The extended period in which animals take a deep sleep in the water.



5. Four-chambered hearts are shared between mammals and which other animal?

6. They are semi-aquatic fish-eating mammals that are relatives of weasels and badgers.



7. These mammals have incisors that can grow to about a metre in length.



8. These animals feed on dead animals and plant materials.



9. These mammals keep their young in a pouch.



Helpbox



Marsupials, Walrus, Otter, Scavengers, Birds, Red Kangaroo, Hibernation, Milk, Skull



Most mammals part to avoid getting too hot. Humans and some other mammals mainly get rid of heat by sweating.



Activity

Find out different types of mammals. Discuss in class.



CHAPTER 3 Volcanoes Around the World

A volcano is a rupture on the crust of a planetary mass project, such as the Earth, which allows hot lava, volcanic ash, and gases to escape from a magma chamber below the surface.

Test your knowledge about volcanoes and answer these questions.

1. What is the life span of land-based volcano?
2. How many volcanic eruptions have occurred over the last 5,000 years?
3. How many of the world's volcanoes are considered potentially active?
4. Most of the volcanoes in the world are located in
5. How hot is lava?
6. The "Ring of Fire" is
7. This was a nine year eruption that destroyed the town. It was in Mexico.
8. What was the largest volcanic eruption (in terms of lava emissions) in modern times?
9. What is the largest volcano in the world?
10. What's the only active volcano on the European mainland?

Scientists use the term Peleem to describe the most violent kind of volcanos and Hawaiian for the least violent volcanos.



Activity

Find out more on volcanos. Prepare a presentation on it.



CHAPTER

Fungi Kingdom

Choose the correct answers to the questions.

- The fungi are now not considered plants as they once were but are part of the :
(a) Eukaryotes (b) Angiosperms (c) Gymnosperms
- Which of these are not in the fungi group?
(a) Mushrooms (b) Angiosperms (c) Yeasts
- A living organism that lives of another living organism without helping that organism is a :
(a) Angiosperm (b) Parasite (c) Gymnosperm
- The structures that fungi have for taking in nutrients are called :
(a) Mycelium (b) Spores (c) Eukaryo
- The form of reproduction in which an organism grows a new piece that eventually breaks off to form a new organism is called :
(a) Fission (b) Budding (c) Spores
- How many species of fungi are thought to exist?
(a) 1,50,000 (b) 1.5 million (c) 1.5 billion
- Fungi made their own food.
(a) True (b) False (c) Don't Know
- Fungi have true roots, stems and leaf structures.
(a) False (b) True (c) Don't Know



Fungi acquire nutrients by absorbing-digesting food outside its body.



Activity

Discuss in group about fungi and its importance.



CHAPTER 5 Unusual Animals

Identify these rare and amazing animals. Pick your answers from the help box.



1. This is the Galucus Atlanticus, a crazy species of blue sea slug. It is able to float on the water's surface because of a gas-filled sac in its stomach. This species can be found in the warm ocean waters, so be careful not to step on these beauties.
2. This frightening-looking fish is found near the Galapagos Islands, but it doesn't swim. It can be found on the BOTTOM of the ocean, walking on its unusual fins.
3. It lives in the trees of the tropical forests in Africa and it does most of its hunting at night. This snake may look fierce, but it is actually one of the smallest species of snake, with a maximum length of only 78 cm.
4. This funny fish in average 30-75 cm in length. Some can grow up to 1.2 m probably because it spends up to 80% of its time looking for food.
5. This rare species of frog commonly found in India is much more bloated and round than its other frog counterparts. It also has a small head and an unusual snout. It is so rare because it only spends two weeks above ground every year.

Helpbox



The Bush Viper, The Red-lipped Batfish, The Blue Parrot Fish, The Blue Dragon, The Indian Purple Frog



The Venezuelan Poodle Moth was only discovered in 2009. They are found in Venezuela and not much is known about these fuzzy, dog-like moths just yet.



Activity

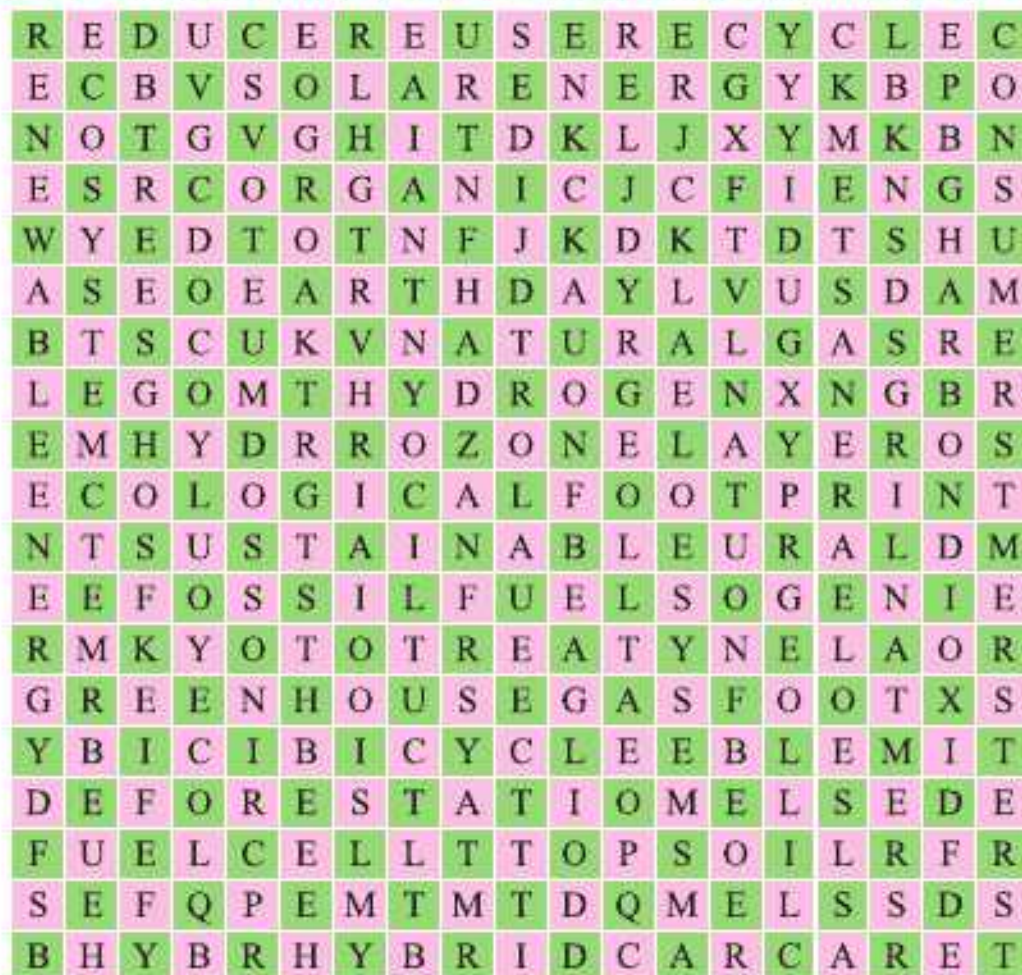
Prepare a presentation on unusual birds.



CHAPTER

Greener Earth

We all deserve a greener, cleaner Earth. Some environment-related words, explained in each clue below, are hidden in this word grid. Can you find them and circle them?



1. A kind of energy that can be used and reused any number of times. _____
2. An electrochemical cell that produces electricity from a replenishable fuel tank by combining hydrogen and oxygen. It can be used in cars. _____
3. An efficient, environment-friendly mode of transport with two wheels. _____
4. A mixture of flammable gases, mainly the hydrocarbons, methane and ethane, that occurs beneath the surface of the Earth. _____

5. Started in 1970, this day is celebrated all over the world on 22 April every year.



6. Food that has been produced without using synthetic fertilizers, chemicals, or hormones.

7. Surface soil, usually including the organic layer in which plants have most of their roots. Over time, it is washed away because of erosion or intensive agriculture.

8. Part of the Earth's atmosphere that repels the sun's ultraviolet rays.



9. The 3R's that can help us lessen our harmful impact on the environment and create a more sustainable world.

10. An international treaty that gave countries a timeline to reduce their emissions of greenhouse gases.

11. The name given to gases that trap heat and cause global warming.

12. One of the most abundant elements in the universe which can be used as a fuel source.

13. A resource that can be used such that it is not depleted or exhausted permanently.



14. The most abundant energy source that can be harnessed by human being.



Corona Virus

Read and learn the following question answer:

1. What is the official name of the disease cause by the novel coronavirus?

Ans. The official name of the disease caused by the novel coronavirus is "Coronavirus Disease 2019" or "Covid – 19" for short.

2. How is the virus that cause Covid – 19 officially designated?

Ans. The virus that cause Covid – 19 is officially designated as "Severe Acute Respiratory Syndrome Coronavirus" or "SARS – COV – 2" for short.

3. What is the taxonomic classification of the virus that cause Covid – 19?

Ans. SARS – COV – 2 is a member of the family coronavirus sub family orthocoronavirinae, genus Betacoronavirus.

4. What are the common symptoms of Covid – 19?

Ans. Common symptoms of Covid – 19 include fever, cough, shortness of breath, loss of taste and smell, and body aches.

5. How is the virus that causes Covid – 19 transmitted?

Ans. The virus that cause Covid – 19 is primarily transmitted through respiratory droplet when an infected person cough or sneezes. It can also be transmitted by touching a surface or object contaminated with the virus and then touching one's face.

6. How does the virus that causes Covid – 19 differ from other Corona Viruses?

Ans. SARS – Co V–2 is a novel corona viruses, meaning. It has not been previously identified in humans is the most closely related to bat coronaviruses but has several genetic mutations that distinguished it from other Coronaviruses.

7. What is the official disease code for Covid – 19 according to the International classification of Disease (ICD)?

Ans. The official disease code for Covid – 19 according to the International classification of Diseases (ICD) is "U07.1"

8. What is the genetic material of SARS – Co V – 2?

Ans. The genetic material of SARS – Co V – 2 is RNA (ribonucleic acid).

9. How does SARS – Co V – 2 enter host cells?

Ans. SARS – Co V – 2 enters host cells by binding to the ACE 2 receptor on the surface of the cell, and then using the spike protein to fuse with the cell membrane.

10. What are the structural proteins of SARS – Co V – 2?

Ans. The structural proteins of SARS – Co V – 2 include the spike (s), protein, envelope (e) protein, Membrane (M) protein, and nucleocapsid (N) protein.

11. How does SARS – Co V – 2 replicate within host cells?

Ans. Once inside the host cell, SARS – Co V – 2 uses the host cell's machinery to replicate its genetic material and produce new viral particles.

12. What are the functions of the spike protein in SARS – Co V – 2?

Ans. The spike protein of SARS – Co V – 2 binds to the ACE 2 receptor on host cells, allowing the virus to enter and infect the cell.

13. How does the human immune system respond to SARS – Co V – 2?

Ans. The human immune system responds to SARS – Co V – 2 by producing antibodies and activating T – cells to target and neutralize the virus.

14. What are the genetic variations of SARS – Co V – 2?

Ans. SARS – Co V – 2 has several genetic variations that have emerged, including the Alpha, Beta, Gamma, Delta and Omicron variants.

15. How does the genetic variation of SARS – Co V – 2 affect the disease course?

Ans. The genetic variations of SARS – Co V – 2 can affect the severity of the diseases, the efficacy of vaccines, and the ability of the virus to evade the immune system.

16. How does SARS – Co V – 2 evade the host immunity?

Ans. SARS – Co V – 2 can evade the host immunity by mutating its spike protein, which allows it to bind to the ACE 2 receptor more efficiently and avoid recognition by antibodies.

17. How does the genetic variation of SARS – Co V – 2 affect the diagnostic test?

Ans. The genetic variation of SARS – Co V – 2 may affect the diagnostic tests by making them less sensitive or specific. This is why diagnostic tests need to be updated frequently to account for the genetic variations.

18. What are SARS – Co V – 2 variatints?

Ans. SARS – Co V – 2 variants are different genetic form of the virus the causes Covid – 19. They can have changes in their spike protein and other genetic regions that can affect the virus, transmissibility, viruences, and vaccine efficiency.

19. How do SARS – Co V – 2 variants emerge?

Ans. SARS – Co V – 2 variants emerge as a result of genetic mutation that occurs as the virus replicates in human hosts. These, mutations can accumulate over time, leading to the emergence of distinct genetic forms of the virus.

20. How do scientist identify SARS – Co V – 2 variants?

Ans. Scientists identify SARS – Co V – 2 variants by sequencing the viral genome of the samples, and then comparing the genetic sequences to those of known variants.

21. What are some examples of SARS – Co V – 2 variants?

Ans. Some examples of SARS – Co V – 2 variants include Alpha, Beta, Gama, Delta and Omicron variants.

22. How does the Alpha variants of SARS – Co V – 2 differ from the original strain?

Ans. The Alpha variants of SARS – Co V – 2, also known as B. 1.7 has a mutation in the spike protein that allows it to bind to the ACE 2 receptor more efficiently and increase its transmissibility.





CHAPTER 8 International System of Units

A unit is a definite magnitude of a physical quantity, defined and adopted by convention or by law, that is used as a standard for measurement of the same physical quantity.

Write the correct letters in the boxes to match each of these units with what they are used to measure.

- | | | |
|---------------|--|---|
| 1. Ampere | | (a) Energy |
| 2. Candela | | (b) Power |
| 3. Light-year | | (c) Voltage |
| 4. Newton | | (d) Electric Current |
| 5. Joule | | (e) Resistance |
| 6. Ton | | (f) Plane Angles |
| 7. Ohm | | (g) Force |
| 8. Volt | | (h) Weight |
| 9. Watt | | (i) Distance travelled by light in a year |
| 10. Radian | | (j) Intensity of light |

The Wheaton is a unit of measuring Twitter Followers! This measurement was created when American actor and writer Wil Wheaton achieved half a million Twitter Followers.



Activity

Find 10 other units of measurement along with their symbols. Discuss in class.



Amazing Facts About the Human Body

The human body is an incredibly complex and intricate system and it still baffles researchers regularly despite thousands of years of medical knowledge. As a result, it shouldn't be a surprise that even body parts we deal with everyday have unexpected facts and explanations behind them.

Here are some wacky facts about the human body.

1. The higher your IQ, the more you dream.
2. The nail on the middle finger grows faster than the other fingernails.
3. Fingernails grow nearly four times faster than toe nails.
4. The life span of a human hair is 3 to 7 years on average.
5. Women's hearts beat faster than men's.
6. Your eyes remain the same size after birth but your nose and ears never stop growing.
7. We are about 1 cm taller in the morning than in the evenings.
8. The colder the room you sleep, the higher the chances are that you would get a nightmare.
9. The only part of your body that has no blood supply is the cornea in the eye. It gets its oxygen directly from air.
10. Everybody has one strong eye and one weak eye.
11. The strongest muscle in the body is the human tongue.
12. The hardest bone in the human body is the jaw bone.
13. It takes the body around 12 hours to completely digest eaten.
14. Your sense of smell is around 10000 times more sensitive than your sense of taste.
15. Antibiotics are only effective against bacteria, they won't help in fighting off a virus.



QUIZ TIME

1. Blood contains red blood cells, white blood cells and _____.
2. The _____ stores the bile used to break down dietary fat.
3. Hair is made mostly of a protein called _____.
4. The study of the human heart and its various disorders is known as _____.
5. The nervous system can _____ signals to activate muscles or glands.
6. The five main sense organs are your eyes, ears, nose, _____ and _____.
7. The longest bone in the human body is the thigh bone called the _____.
8. The human tongue is divided into two parts—the _____ and the _____.
9. _____ can damage a teeth if left untreated.



At birth the human skeleton is made up of around 300 bones. By adulthood, some bones have fused together to end up with 206 bones.



Activity

Find out interesting facts about skin and discuss in class.

10

CHAPTER

Amusing Gadgets

When you go to the supermarket, do you often forget to purchase certain items and remember them only when you are back home? The **Smart Shopper Automated Grocery List Printer** can be of great help in such a situation. It is a device that is voice activated. When you talk into the handset, it compiles a textual list ready for printing. You can stick the list with a magnet on your fridge. The list is stored by types of items that one would need to buy, to avoid wasting time browsing, and comes pre-programmed with 2500 items, with the option to add your own. Ink refills are not required in this machine.



There is no better alarm system than an angry, protective dog barking ferociously, but not all of us are able to keep a dog that would guard our houses. So a barking protection system has been invented, called an **Alarm Security Bar**, which can add instant protection to any home. It does not need any expensive installation and monitoring. The alarm is turned ON whenever the door knob is touched or the door is pushed, and sets OFF under the slightest of pressures. The alarm can either be set to a powerful siren or to the alternative barking alarm sound.

If an alarm clock cannot help you wake up on time because you always end up hitting the snooze button, you need the **Flying Alarm Clock**. The idea is simple-when the alarm is triggered, part of the alarm will fly with the help of a propeller and land at some random location in your room. Only when the piece is placed back on the stationary alarm piece, will the alarm cease. By then, you will be forced out of your bed.



If you hate being outdoors in hot summers, here is an invention for you! The **Intelli Cool** is said to be the first ever outdoor air conditioning unit that uses a special water evaporation technology. The fans attached to it will cool the air in nearby surroundings. It also provides bright light for the hot sunny days.

The **LED Umbrella** comes equipped with a built-in LED light which illuminates in the dark. At the touch of a button, the light embedded inside the handle shines to lighten the area under the umbrella hood. So for those of you who feel uncomfortable about taking a long walk home in the dark, this LED umbrella is sure to make that walk a little bit brighter.



QUIZ TIME

1. Who invented the Nuvaring Birth Control?
2. Name the invention by Doug Engelbart in the 1960s which he called 'x-y indicator'.
3. What is the name of the manufacturing company that makes Abiocr Artificial Hearts?
4. In 2004, which company invented the 'thinking shoes', with a built-in microprocessor that decides whether the wearer needs soft or firm support?
5. For whom was the global positioning system (GPS) developed in the 1970s?



The first vending machine was invented in 62 CE.



Activity

Do you know what Chindogu is? Find out.



Study of Robots

Robotics is the branch of mechanical engineering, electrical engineering and computer science that deals with the design, construction, operation and application of Robots.

You must have seen robots doing amazing things in movie and on television or have read about them in books. But what is a robot? A robot can be defined as a machine that looks like a human being and performs various complex acts like a human being.

The famous Sci-fi writer Isaac Asimov was one of the first people to come up with the idea of a fictional friendly robot, governed by three laws that he invented in his story Runaround. According to his laws :

- ❑ A robot may not injure a human being, or, through inaction, allow a human being to come to harm.
- ❑ A robot must obey orders given by human beings, except where such orders would conflict with the first law.
- ❑ A robot must protect its own existence as long as such protection does not conflict with the first or second laws.

Name the famous writers or film makers who created these robots.

1. R2D2 _____
2. Sonny _____
3. The Sentinels _____
4. Chitti _____



Leonardo da Vinci designed a robot, and probably built one too around 1495! It was a robotic knight with a complicated array of gears and pulleys that could sit up, wave its arms, and move its head.



Activity

If you could have a robot that would be your genie and do any task for you, what would you want it to do?



Study of Genes

A gene is the molecular unit of heredity of a living organism.

Do you and your mother have the same eye colour? Is your hair as curly as your father's? This similarity may be explained by genetics. Genetics is the study of heredity, or how certain features pass from parents to their young ones.

A. Write 'T' for the sentences which are true and 'F' for those which are false.

1. Austrian botanist Gregor Mendel studied garden pea plant to establish his laws of inheritance.
2. The cell is the smallest self-contained living unit.
3. In genetics, cloning is the process of creating an organism that is an exact replica of another.
4. Humans usually have 48 chromosomes.
5. All genes of a person are equally dominant.
6. Genes exist in our cells.



DNA is stored in the cell nucleus. Its structure is like a twisted ladder with four different types of rings. The rings make up a four-letter alphabet that spells out your genes, like letters in a book.

Genetic Glossary :

Base — A building block of DNA

Gene mapping — Locating the positions of genes in a Chromosome

Mutation — An inheritable change resulting from an alternation in DNA

B. Fill in the blanks. Choose your answers from the help box.

1. _____ are unique to each person, even identical twins.
2. _____ was the first cloned sheep.
3. DNA, one of the largest known molecules, consists of two long thin strands that are wound around each other to form a spiral. This special structure is called a _____.
4. A chromosome is a molecule that controls _____.
5. In the nucleus of each cell, the DNA molecule is packaged into thread-like structures called _____.
6. Unless you have an _____ twin, your DNA is unique.
7. _____ is stored in the cell nucleus.
8. _____ is a liquid tissue.
9. Cells of the same type group together to form _____.

Helpbox



Chromosomes, Dolly, Blood, Tissues, Identical, Heredity, Finger Prints, Double Helix, DNA



Humans are 99.9% genetically identical—only 0.1% of our genetic make-up differs.

13

CHAPTER

History of Computers

The computer as we know it today had its beginning with a 19th century English mathematics professor name Charles Babbage. He designed the Analytical Engine and it was this design that the basic framework of the computers of today are based on.

Try this quiz to brush up your knowledge of the world of computers. Choose the correct answers.

1. Which American computer company is called Big Blue?

(a) Microsoft

(b) IBM

(c) Compaq Corp



2. When was the first Mini computer built?

(a) 1965

(b) 1962

(c) 1971

3. Which was the most popular first generation computer?

(a) IBM 1650

(b) IBM 360

(c) IBM 1130



4. When did IBM introduced the 80286 based PC/AT?

(a) 1982

(b) 1984

(c) 1985

5. Who is credited with the idea of using Punch Card to control patterns in a weaving machine?

(a) Hollerith

(b) Babbage

(c) Joseph Marie Jacquard

6. Who invented the Microprocessor?

(a) Marican E Huff

(b) Hollerith

(c) Pascal

7. When was the worlds first laptop introduced in market and by which company?

(a) Epson, 1981

(b) IBM, 1950

(c) Microsoft, 1988

8. Which was the world first mini computer?

(a) PDP-1

(b) IBM system/36

(c) PDP-II

9. First digital computer built with IC chips was known as :
 (a) IBM 7090 (b) Apple-1 (c) IBM system/360
10. In which year did Motorola come out with its enhanced 68030 processor?
 (a) 1964 (b) 1985 (c) 1986
11. In which year was the first computer mouse invented?
 (a) 1980 (b) 1963 (c) 1958
12. What was the memory storage capacity of the first hard drive?
 (a) 100 MB (b) 5 MB (c) 75 MB
13. Which two numbers together form the 'binary language' used in computers?
 (a) 0 and 9 (b) 1 and 3 (c) 0 and 1
14. What is the name of India's first indigenously designed super computer?
 (a) Kalki (b) Pushpaka (c) Param
15. MPG extension refers usually to what kind of file?
 (a) MS office document (b) Animation/movie file
 (c) Image file
16. What do the letters MM stand for in the abbreviation MMS?
 (a) Messaging Media (b) Multimedia Messaging
 (c) Multiple Messaging



1. TYPEWRITER is the longest word that can be made using the letters only on one row of the keyboard.
2. While it took the radio 38 years, and the television a short 13 years, it took the World Wide only 4 years to reach 50 million users.



Activity

What do you know about the information superhighway? Make a presentation on it.

14

CHAPTER

Nuclear Energy

Nuclear energy is the use of exothermic nuclear processes, to generate useful heat, electricity or other radiation.

Answer these questions with the help of help box.

1. The number of protons in a nucleus is the same as the number of _____ surrounding it, thus making the atom electrically neutral.
2. In 1905, Albert Einstein published his _____ from which he derived the famous equation : $E = mc^2$.
3. The atom bomb is based on which nuclear principle? _____
4. Electrons are not the smallest particles present in atoms. Electrons are always made of smaller particles called _____.
5. At the centre of each atom is a tiny core called a _____ which is surrounded by a cloud of electrons or electrically charged particles that move in the outer regions of the atom.
6. The mass of a nucleus is slightly less than the sum of its constituent neutron and proton masses. This so-called mass defect is also known as the _____.



Helpbox



Binding energy, Electrons, Quarks, Theory of Relativity, Nuclear Fission, Nucleus



A nuclear power reactor generates heat that is converted into steam. The steam can be used directly for power, as in a nuclear submarine. It can also be used to generate electric power.



Activity

Find out about five nuclear power plants. Discuss in class.



Diseases

A disease is a particular abnormal, pathological condition that affects part or all of an organism. Although medical science has made great progress in eradicating diseases, many ailments remain unconquered. Nevertheless, research is underway against these and other baffing diseases in the hope that some day they will be wiped out or, at least, be manageable.

Choose the correct answers to these questions about diseases.

1. This disease is caused by a bacterium that lives in water. It is common in water with lots of algae.
(a) Dengue (b) Cholera (c) Typhoid
2. Influenza is a serious infection of the upper or lower respiratory tract. Common symptoms of influenza include fever, chills, weakness and pain in the muscles, along with occasional soreness in the throat and abdomen. What causes influenza?
(a) Fungus (b) Virus (c) Bacteria
3. Insulin helps to move sugar from the blood into the cells, where it can be used to get energy. A lack of insulin or an inability of the body to utilize the insulin that is produced by the body may cause diabetes mellitus. Diabetes affects the :
(a) Gall Bladder (b) Pancreas (c) Kidneys
4. This disease is an eating disorder affecting persons who think that they are overweight and tend to eat less and less, continuing to lose weight until they are malnourished :
(a) Anotia (b) Syphilis (c) Anorexia
5. Alzheimer's disease leads to severe memory loss and a gradual loss of the mental faculties. It is a major cause of dementia, which generally occurs in old age. It is an incurable, degenerative disease of the :
(a) Spine (b) Brain (c) Liver
6. It is an infectious disease nowadays termed as Hansen's disease. It is a mildly infectious disease and is prevalent in areas where there are high levels of poverty.
(a) Leprosy (b) Herpes (c) HIV

7. The disease called Acquired Immunodeficiency Syndrome, or AIDS, is a complicated illness that damages the human body's immune system, or the system responsible for fighting diseases, and makes the victim vulnerable to various infections. The virus that causes AIDS is called :
- (a) HBV (b) H2N1 (c) HIV
8. This condition, contrary to popular belief, is caught from exposure to infected people, and not from a cold environment, chilled wet feet, or drafts. People can carry the virus and communicate it without experiencing any of the symptoms themselves. Symptoms may include sneezing, headaches, fatigue, chills, sore and nasal discharge etc, which is usually the first warning of the condition. The usual duration of the illness is about five to seven days. Which illness is this?
- (a) Conjunctivitis (b) Common Cold (c) Typhoid
9. How do we commonly know the disease infantile paralysis, which is an infectious viral disease that usually causes mild illness, but when the virus attacks the central nervous system, it may lead to extensive and permanent muscle paralysis, usually in the legs and lower torso?
- (a) Polio (b) Epilepsy (c) Multiple Sclerosis



According to WHO, cancer is a leading cause of death worldwide. About 10 million new cases of cancer are diagnosed around the world every year. 4 February, 2009 had been designated as the WHO World Cancer Day.



Activity

Learn more about the cancer and find how it can be prevented.

16

CHAPTER

Humans Caused Disasters

For centuries, many acts by human being have been the cause of great tragedies. Read about some of the deadliest disasters caused by humans.

Terrorism has been a leading cause of disaster in the past few decades, affecting lives worldwide. One of the ugliest instances of terrorism were the aircraft attacks on the **World Trade Centre**, New York City, and on the Pentagon on 11 September, 2001, causing more than 3,000 deaths. In another instance of terrorism, over 100 people died and more than 300 were injured in the terrorist mayhem in Mumbai on 26 November, 2008. Other than the loss of lives and the resulting agony, terrorism also affects the world financial markets and economies, the relationships between countries and society.



In 1929, the unequal distribution of wealth among people and risky buying and selling of shares in the stock market led to a financial disaster, called the **Great Depression**, which lasted for a decade. It was a severe depression, experienced by the industrialized Western World. Although it originated in the United States, the Great Depression led to severe unemployment worldwide and affected the economies of almost every country. In 2008 the world economy faced another major

financial crisis that began in the US sectors of investment banking and mortgage loans and then spread to financial markets overseas, causing a deep recession worldwide.

On 26 April, 1986, a nuclear power station at **Chernobyl**, then a part of USSR, exploded when a poorly designed experiment was conducted. A large amount of radioactive material was released into the atmosphere causing numerous deaths. Many persons contracted serious radiation sickness. The gas was spread by the wind over parts of USSR and reached as far west as France and Italy. Millions of acres of forest and farmland were contaminated. In the subsequent years, much livestock were born deformed, and several thousands of radiation-induced illness and cancer deaths occurred. The release of toxic gas from the pesticide plant at **Bhopal**, India, in 1984 is another industrial accident. An estimated 10,000 died within 72 hours, and 25,000 since then.





A **famine** is an extreme and long-term shortage of food. It can affect an entire country, as it did in Somalia in the early 1990s. Or it may be regional, as in parts of Ethiopia during the 1980s. In Germany, just after World War II, some areas were devastated by the war and experienced a regional famine.

QUIZ TIME

1. Towards the end of World War I, an epidemic spread across the world claiming 21.5 million lives. Name it.
2. In 1964, due to an unfavourable ruling by the referee, the deadliest soccer riot in history occurred in which country?
3. After a dam burst, which river in India changed its course, caused severe flooding in August, 2008?



The Bengal Famine of 1770 swept away one-third of the population of Bengal. Two more catastrophic famines hit Bengal and Bangladesh in 1943 and in 1974 respectively.



Activity

What is the United Nations Comprehensive Test-Ban Treaty? Find out.



CHAPTER

Quiz on G-20

Tick (✓) the correct answer:

1. When was G20 established?
(a) 1995 (b) 1999 (c) 1985 (d) 2000
2. Which of the following cities is organizing the recent G20 summit going to be held in 2023?
(a) Brisbane (b) Osaka (c) Antalya (d) India
3. Which of the following is not the member of the G20?
(a) Indonesia (b) Singapore (c) Mexico (d) Turkey
4. Where was the first summit of G20 held?
(a) USA (b) Britain (c) Canada (d) France
5. To date how many summits of G20 summit?
(a) 5 (b) 9 (c) 11 (d) 18
6. When will India hold the G20 summit?
(a) 2020 (b) 2023 (c) 2022 (d) None of these
7. Where is the headquarters of G20?
(a) India (b) USA (c) Turkey (d) None of these
8. What is the official language of the G20 summit?
(a) English (b) French (c) Spanish (d) All of the about
9. Which country will host G20 summit in 2024?
(a) South Africa (b) Brazil (c) Russia (d) Japan
10. PM Modi transferred the G20 presidency to:
(a) Carmen Lucia (b) Jair Bolsonaro
(c) Luis Inaciolula Da Silva (d) Michel Temer
11. What is the meaning of G20?
(a) Grand 20 nations (b) Great 20 countries
(c) Group of 20 (d) None of about
12. What is the inspiration for G20 2023 logo?
(a) Indian National flag (b) Russian's National flag
(c) French's National flag (d) None of the above



Know About India

Tick (✓) the correct answer:

1. What is the theme of International Olympics day 2024?
(a) Together, for a peaceful world ☐ (b) Let's move and celebrate ☐
(c) Together for a better world ☐ (d) Moving forward: United by Emotion ☐
2. Recently, who has become first Indian male wrestler to qualify for 2024 a Paris Olympics?
(a) Bhim singh ☐ (b) Aman Sehrawat ☐
(c) Dinkarrav Shinde ☐ (d) Udey Chand ☐
3. Recently, who achieved India's first rowing quota for a Paris Olympics?
(a) Balraj Panwar ☐ (b) Sawarn Singh ☐
(c) Arjun Lal ☐ (d) Jaswinder Singh ☐
4. Who has become the first Indian to be appointed a jury member at the 33rd a Summer Olympics 2024 to be held in Paris?
(a) Bilquis Mir ☐ (b) Madhumita Bisht ☐
(c) Rajyalakshmi Singh ☐ (d) Sania Mirza ☐
5. Recently, who has become the sole Indian weightlifter who qualified for the Paris Olympics 2024?
(a) Mirabai Chanu ☐ (b) Kunjarani Devi ☐
(c) Grudeep Singh ☐ (d) Karnam Malleswari ☐
6. Recently, who has been appointed as the flag bearer of the Indian contingent at the Paris Olympics 2024?
(a) Sathiyana Fnanasekaran ☐ (b) Sharath Kamal ☐
(c) Ankita Das ☐ (d) Soumyajit Ghosh ☐
7. Who has been selected as torch bearer for the 2024 Paris Olympics?
(a) Saina Nehwal ☐ (b) Neeraj chopra ☐
(c) Abhinav Bendra ☐ (d) PV sindhu ☐

8. Who recently secured the India's 16th quota for Paris Olympics?
- (a) Rhythm Sangwan (b) Jiin Yang
- (c) Koushik (d) Yeji Kim
9. Which country has announced to become the first European nation to issue digital Schengen visas for visitors to the 2024 Paris Olympics?
- (a) Switzerland (b) France
- (c) Germany (d) America
10. Special Olympics Bharat (SOB), recently seen in news, is a national sports Federation recognized by which ministry?
- (a) Ministry of Youth Affairs and Sports
- (b) Ministry of Home Affairs
- (c) Ministry of Power
- (d) Ministry of Defence





Election Commission of India

Election commission of India submits list of newly elected members to the 18th Lok Sabha to the Hon'ble President of India.

Chief Election Commissioner Shri Rajiv Kumar along with Election commissioners Shri Gyanesh Kumar and Dr Sukhbir Singh Sandhu, called on the Hon'ble President of India at 1630 hrs today (June 6, 2024). A copy of the notification issued by the election commission of India in terms of Section 73 of the Representative of people act 1951, containing the names of elected members to the House of people, following General Elections to the 18th Lok Sabha was submitted by them to the Hon'ble President.

Afterwards, chief Election commissioner, both the Election commissioners and senior officers of the commission went to Rajghat to seek blessings of the father of the Nation after successful conduct of the General Elections 2024.



20

CHAPTER



UNIT 4

Sports and Entertainment

Director Watch

A Film Director is a person who directs the making of a film, a film director controls a film's artistic and dramatic aspects, and visualizes the script while guiding the technical crew and actors in the fulfillment of that vision.

These famous directors are known for their style of film making. Can you name them?

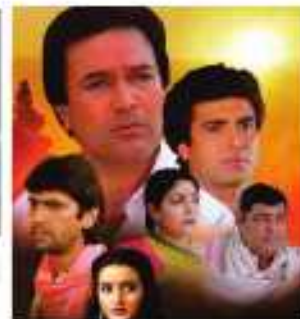
1. This Toronto-based director has earned international attention with her films. She is considered by many as one of the finest new directors on the horizon. With films such as the trilogy that consists of Fire, Earth and Water. She is quickly becoming the voice of a new India.



2. Labelled the eternal romantic and with one of the best musical senses in the business, he is arguably India's most successful director of romantic films. Although he made action-oriented films like the ever popular Deewar, it is in taking love and its various aspects that he has been at



3. Considered to be a man ahead of his time, he was one of the greatest icons of commercial Indian Cinema. Although he made less than 50 films during his life time, they are believed to be the best to come from Bollywood's Golden Age, known both their ability to reach out to the common man and for their artistic and lyrical content.





4. He was the son of well known Indian actor Prithviraj Kapoor, who acted both in film and on stage. After apprenticing in the Bollywood Production Studios of the 1940's at 24 years of age he produced, directed and acted in Aag, with his new company, RK films.



5. Accomplished film Director/Writer/Producer, she was born in India and educated at Delhi University and at Harvard. She began her film career as an actor and then turned to directing award winning documentaries, including so far from India and India Cabaret. Her debut feature film was, Salaam Bombay.



At the age of 11, Kishan Shrikanth from India is perhaps the youngest film director ever! His first film Footpath, about children living on India's city streets, has travelled to many prestigious film festivals.



Activity

Find out why film has often been called a director's medium.



Classic Country Music Favourites

Choose the correct option.

- Which word complete the title of this Johnny Cash song? "Sunday Morning Coming _____"?
(a) Fast (b) Home
(c) Down (d) From Texas
- Do you recall the Jeanne C. Riley's hit which contains these lyrics: "I can still recall the word she had to say, she said I'd like to address this meeting of the _____"?
(a) Harper Valley P.T.A. (b) S.P.C.A.
(c) N.R.A. (d) D.C. Highway
- Which word completes the classic lyric: "There's a tear in my beer Cause I'm crying for you _____"?
(a) Here (b) Sweetheart
(c) Dear (d) Claire
- Complete the line to this popular song: "I'm proud to be an Okie from _____".
(a) The hills of Tennessee (b) The pokie
(c) Muskogee (d) The hills of Magee
- Which state completes the line to this song: "Let's go to Luckenbach _____"?
(a) Tennessee (b) Arkansas
(c) Texas (d) Mississippi
- Can you complete the lyric to this Johnny Cash hit: "I'm stuck in Folsom _____"?
(a) County (b) Prison
(c) Texas (d) Jail
- Can you recall the line that completes these lyrics: "I put a nickel in the jukebox. And played the _____"?
(a) Long lonesome road (b) Old fashioned song
(c) Truck driving man (d) Dancing song

22

CHAPTER Musicians

A musician is a person who is talented in making music or performing music creatively, or one who composes, conducts, or performs music.

There are some of the best known musicians of the world. Identify them. You can choose your answers from the help box.

1. He often hailed as the godfather of world music and ambassador of Indian music all across the globe. He composed in 1970 'Concerto For Sitar and Orchestra'. For his extraordinary contribution to music, he was also awarded the highest civilian honour of France i.e. 'Knight of the Legion of Honor'.
2. This composer and child prodigy from Vienna is often considered the greatest musical genius of all time. In his short life, he worked on 16 operas, 41 symphonies, 27 piano concertos, and 5 violin concerti, and other works in every form popular during his time.
3. He is one of the most well-known Indian flute players today. He was awarded the Padma Bhushan in 1992 and the Padma Vibhushan in 2000.
4. This musical giant from Germany successfully combined German, French, Italian, and English musical styles in about 40 operas, 20 oratorios and numerous other vocal pieces, instrumental works, and church music.
5. A well-known Russian composer, he wrote some of the most popular concerts and theatrical music, including the ballets Swan Lake, The Sleeping Beauty, and The Nut Cracker.

Miyan Tansen, one of the greatest musicians of India, was so melodious and powerful that it is said he could summon the rains with his Raga Megh Malhar, and light lamps singing Raga Deepak.



Helpbox



Pyotr Tchaikovsky, Ravi Shankar, Georg Friedrich Handel, Mozart, Hariprasad Chaurasia



Activity

Prepare a Presentation on Indian Musicians.

23

CHAPTER

Digital Photography

Digital photography uses an array of electronic photodetectors to capture the image focused by the lens, as opposed to an exposure on photographic film.

Try this quiz to brush up your knowledge of the world of digital photography.

1. Lenses are usually coated with :

(a) Magnesium fluoride



(b) Sodium fluoride



(c) Zinc fluoride



(d) Copper fluoride



2. 'Camera' is actually a Latin word meaning :

(a) Picture box



(b) Room



(c) Light box



(d) Dark room



3. Who coined the term 'photography'?

(a) Fox Talbot



(b) Nicéphore Niépce



(c) Sir John Herschel



(d) Ansel Adams



4. A small camera, the kind that can fit in your pocket :

(a) Disposable



(b) Point and shoot



(c) DSSR



(d) DSLR



5. Electronic flash memory data storage device used for storing digital information:

(a) Flash drive



(b) Flash card



(c) Memory card



(d) Tripod



6. The part of the camera that indicates the field of view that is seen by the camera lens.

(a) View finder



(b) LCD display



(c) Lens



(d) Camera body



7. The total amount of light that is captured :

(a) Focus



(b) Aperture



(c) Shutter speed



(d) Exposure



8. This means that your subject is sharp and not blurry :

(a) Image noise	☐	(b) Focus	☐
(c) Framing	☐	(d) Exposure	☐
9. The distance from the centre of a lens to the focal point :

(a) Focal length	☐	(b) Auto focus	☐
(c) Depth of field	☐	(d) Aperture	☐
10. The button at the top of the camera used to take the picture :

(a) Camera body	☐	(b) Shutter speed	☐
(c) Shutter release	☐	(d) Lens	☐
11. A camera lens that magnifies the image :

(a) Zoom lens	☐	(b) Exposure	☐
(c) LCD Display	☐	(d) Auto Focus	☐
12. An adjustable opening in a camera that limits amount of light :

(a) ISO Rating	☐	(b) Shutter speed	☐
(c) Shutter release	☐	(d) Aperture	☐
13. A three-footed stand often used to balance a camera :

(a) Flash drive	☐	(b) Tripod	☐
(c) Memory card	☐	(d) Filter	☐
14. The portion of a scene that appears acceptably sharp in the image :

(a) Shutter speed	☐	(b) Aperture	☐
(c) Depth of field	☐	(d) ISO Rating	☐



The largest collection of cameras in the world is held by Dilish Parekh of Mumbai. He has a collection of 4,425 antique cameras.



Activity

Find out about famous photographers of the world. Discuss in class.



Hollywood and Bollywood

Hollywood and Bollywood are both popular cinemas and their names resonate their success around the world. Hollywood is known world-wide and Bollywood is popular in the eastern countries and has been gaining popularity in many western countries including the US. Hollywood is a term that has become synonymous with the American cinema, while Bollywood is highly affiliated to Indian cinema.

Try this quiz to brush up your knowledge of the world of Hollywood and Bollywood.

- Which Bollywood movie won the Grand Prize at the first Cannes film festival?
(a) Do Bigha Zamin (b) Neecha Nagar
(c) Baiju Bawra (d) Mughal-e-Azam
- The world's first commercial motion picture exhibition was held in which year?
(a) 1894 (b) 1902
(c) 1915 (d) 1928
- Which was the first movie ever shot in Hollywood?
(a) Mary Pickford (b) Birth of a Nation
(c) In Old California (d) The Jazz Singer
- In which year the first movie was shot in Hollywood?
(a) 1868 (b) 1902
(c) 1881 (d) 1910
- Which Bollywood movie won the film of the year award in first Stardust Awards?
(a) Black (b) Kaante
(c) Jagriti (d) Omkara
- Which was the first movie to be filmed in Technicolor?
(a) Becky Sharp (b) Grass
(c) Grass Sandwich (d) Extase
- Which was the first Hollywood movie to gross over \$ 100 million?
(a) Black Sharp (b) Grass Sandwich
(c) Jaws (d) Extase

8. Which Hollywood movie hit \$ 100 million mark fastest?

(a) Black Sharp		(b) Titanic	
(c) Jaws		(d) The Twilight Saga : New Moon	
9. Which was the first Hollywood movie shot in Cinema Scope?

(a) The Robe		(b) Ghost Smashers	
(c) Down the wire		(d) Toy Story	
10. Which Bollywood movie won the Best Film award in first Bollywood Movie Awards?

(a) Kaho Naa Pyar Hai		(b) Kuch Kuch Hota Hai	
(c) Kabhi Khushi Kabhi Gham		(d) Devdas	
11. In which movie Raj Kapoor launched dream girl, Hema Malini in the lead Role?

(a) Sapno Ka Saudagar		(b) Neela Sagar	
(c) Ashique		(d) Dharti	
12. The first 3D movie in Hindi was :

(a) Jai Santoshi Maa		(b) Bal Krishna	
(c) Krishna		(d) Shiva Ka Insaaf	
13. Which Bollywood movie was the first cinema scope film?

(a) Kisan Kanya		(b) Kagaz Ke Phool	
(c) Raja Harischandra		(d) Alam Ara	



Bollywood was born 11 years before Hollywood! Bollywood's first production was in 1899 short film, whereas Hollywood's first film came out in 1910.



Activity

Find out more on Bollywood. Discuss in class.



CHAPTER

Indian Sports Women

Indian sportswomen includes the champion sportswomen that India has ever produced who brought to their motherland by establishing record and winning titles on international stage.

Answer the following :

SPORTSWOMEN	SPORT	AWARD
1. Anita Sood		
2. Jwala Gutta		
3. Bula Choudhury		
4. Dala Banerjee		
5. Jhulan Goswami		
6. MC Mary Kom		
7. Reeth Abraham		
8. Saina Nehwal		
9. Sania Mirza		
10. Tania Sachdev		



P.T. Usha was the first Indian woman to reach Olympic finals. She reached the finals of the 400 metres hurdles in the 1984 Los Angeles Olympics. Usha lost the bronze by 1/100th of a second.



Activity

Find out more on Indian sportswomen and discuss in group.



Indian Sports Trophies and Awards

In India sports awards are presented by the Government of India to athletes and sports persons for their accomplishments and outstanding performance in the various fields of sports. The awards are also given for the National level competitions and for sports intended for the physically challenged.

Read about reputed and esteemed sports awards in India.

1. **Arjuna Award :** The Arjuna Awards were established by the Indian Government in 1961 for the excellent performance by the sports persons in various events. The award holds a cash prize of INR 5,00,000, a bronze statue of Arjuna and a scroll. As per revised guidelines, a sports person must have a constant excellent performance for the previous three years at international level to be eligible for the award.
2. **Dronacharya Award :** Instituted in 1985, the Dronacharya Award is presented by the Indian Government for brilliance in sports coaching. The award consists of a bronze figurine of Dronacharya, a cash prize of INR 5,00,000 and scroll of honour. Since the best sportsperson award is named Arjuna Award, it would be appropriate that the best coaching award is named Dronacharya, the Guru of Arjuna.
3. **Dhyan Chand Award :** The Dhyan Chand Award is the highest lifetime achievement award in sports and games in India and is presented by the Indian Government. Initiated in 2002, the award bears a cash prize of INR 5,00,000, a figurine, ceremonial dress and a scroll of honour. It is named after renowned Indian hockey player Dhyan Chand.
4. **Rajiv Gandhi Khel Ratna :** In India, the Rajiv Gandhi Khel Ratna Award is the highest honour given for accomplishment in sports. The term “Khel Ratna” translated in Hindi literally means “sports gem”. Named after the former Prime Minister of India Rajiv Gandhi, the award carries a medal, a significant cash prize of INR 7,50,000 and a scroll of honour. The award was initiated in 1991-92 to offer a highest national-tribute in the field of sports. The award was formulated to present the utmost honour for exceptional sporting performance, whether by an individual or a team in all sporting disciplines in a year.

QUIZ TIME

1. Winner of Arjuna Award 2023 in Boxing
2. Winner of Arjuna Award 2023 in Cricket
3. Winner of Arjuna Award 2023 in Wrestling
4. Winner of Dronacharya Award 2023 in Archery
5. Winner of Dronacharya Award 2023 in Wrestling
6. Winner of Dronacharya Award 2023 in Hockey
7. Winner of Dhyan Chand Award 2023 in Athletics
8. Winner of Dhyan Chand Award 2022 in Hockey
9. Winner of Dhyan Chand Award 2023 in Wrestling
10. Winner of Rajiv Gandhi Khel Ratna Award 2023 in Shooting



Sachin Tendulkar receives the Bharat Ratna Award in February 2014. He is the first sportsman and youngest Indian who got Bharat Ratna.



Activity

Find out more about Indian sports trophies.



Hockey Facts

Read about some interesting Hockey facts.

- ❖ London was the first city in England to popularize this game in 1870s.
- ❖ 'Blackheath Rugby and Hockey Club' is the first hockey club in the world which was set up in the year 1861 in England.
- ❖ In the years 1886, Hockey Association of England was formed in London.
- ❖ The apex institution of hockey is 'Federation Internationale de Hockey' which was established on January 7, 1924, with headquarters at Vienna (Austria), which was later shifted to Paris.
- ❖ The first International Hockey Match was played between Wales and Ireland in Rayle on 26 June, 1895.
- ❖ Hockey was introduced in the Olympic Games for the first time in 1908 in London.
- ❖ Indian Hockey Federation (IHF) was formed on 7th November, 1925 at Gwalior.
- ❖ Since 1944, National Hockey Championship is organised by the IHF every year.
- ❖ All India Women's Hockey Federation was formed in 1947.
- ❖ India took part in Olympics for the first time in 1928 (Amsterdam Olympics).
- ❖ In Olympics, India has won the hockey title a maximum of 8 times. The first World Cup Hockey was played in Barcelona in 1971.



SOME IMPORTANT MEASUREMENT IN HOCKEY

No. of Players	—	11 Players in each team
Length of Playing Field	—	91.44 meters
Width of Playing Field	—	50 to 55 meters
Weight of the Ball	—	155 to 163 grams
Circumference of the Ball	—	223–224 cm
Colour of the Ball	—	White
Weight of Hockey Stick	—	280 grams (max.)

QUIZ TIME

Complete the following table based on Hockey World Cup Information.

YEAR	PLACE	WINNER	RUNNER UP	INDIA'S POSITION
1. 1971	_____	Pakistan	_____	Third
2. 1972	Amsterdam	Holland	_____	Second
3. 1975	_____	_____	Pakistan	First
4. 1978	Buenos Aires	_____	_____	Sixth
5. 1982	_____	Pakistan	West Germany	_____
6. 1986	_____	Australia	England	Twelfth
7. 1990	Lahore	_____	_____	Tenth
8. 1994	Sydney	Pakistan	_____	_____
9. 1998	Utrecht	_____	Spain	_____
10. 2002	Kuala Lumpur	Germany	_____	_____
11. 2006	Germany	_____	Australia	Eleventh
12. 2010	New Delhi	_____	Germany	Twelfth
13. 2014	Netherlands	_____	Netherlands	Eighth



28

CHAPTER

Racing Around the World

Have fun with this mixed up racing quiz. Choose the correct answer.

- Who was the 1997 Formula 1 World Champion?
 (a) Michael Schumacher (b) David Coulthard
 (c) Mika Hakkinen (d) Jacques Villeneuve
- Do FIA World rally championship cars make pitstops during a stage?
 (a) Yes (b) No
- How many lugnuts are on a formula 1 car?
 (a) 5 (b) 1 (c) 4 (d) 2
- Which of the following drivers has won both pole position and the race, at the United States Grand Prix?
 (a) David Coulthard (b) Eddie Irvine
 (c) Mika Hakkinen (d) Michael Schumacher
- Who was the first driver to win the Daytona 500, the indy 500 and win the formula championship?
 (a) Mario Andretti (b) Alex Zanardi
 (c) Dan Gurney (d) Juan Montoya
- What famous road racer won a NASCAR race in his third Grand National Street?
 (a) David Bell (b) Kevin Harvick
 (c) Dan Gurney (d) Richard Petty
- How many points do you get for leading a lap in F1 competition?
 (a) 1 (b) 10 (c) 5 (d) 0
- What type of engine did an F1 car run, in 2000?
 (a) Alcohol (b) Normally aspirated
 (c) Highly tuned (d) Jet engines



Approximate 80,000 components come together to make an F1 car. The car has to be assembled with cent percent accuracy. If it were assembled 99.9% correctly, it would go on the track with 80 components wrongly placed.

This is a general quiz about the martial arts. Take the quiz and choose the correct option.

- Who is the father of American Kenpo?
 - Bruce Lee
 - ED Parker
 - Book Sal Wan
 - Chuck Norris
- What is the original style Bruce Lee learned?
 - Wing Chun
 - Akido
 - Wushu
 - Tang Soo Do
- Aikido is from what style?
 - Aiki Jitsu
 - Jujitsu
 - Judo
 - Tatsu do
- Judo is from what style?
 - Wing Chun
 - Akido
 - Jujitsu
 - Kenpo
- The Samurai code of honour is known as _____.
 - Kakido
 - Chendo
 - Bodido
 - Bushido
- What family was involved at the start of the popular fight series the UFC?
 - Blanks
 - Gracies
 - Lees
 - Chongs
- What was the name of movie, Bruce Lee was working on when he died?
 - Game of Death
 - Enter the Dragon
 - Chinese Connection
 - Fists of Fury
- Who played Darth Maul in 'Phantom Menace'?
 - Jackie Chan
 - Ray Park
 - Jet Li
 - Billy Blanks



Activity

Prepare a presentation on five martial artists and discuss in class.

Jackie Chan broke twelve concrete blocks with his hand, while keeping the egg he was holding intact.





CHAPTER

World Athletics Championship

It was a historic year for team USA track and field in Budapest as Noah Lyles won the print double, Sha'carri Richardson took the 100m title and the men's and women's 4 × 100m relays swept the gold medals for the first time since 2007. Lyles was also the first American man to win the sprint double since Tyson Gay 16 years prior.

The United States dominated the medal table, taking 29 medals, including 12 golds ahead of second place Canada's four.

Sha'carri Richardson of team United States celebrates with Christian Coleman, Noah Lyles and Fred Kerley after winning the women's 4 × 100 m Relay final during day eight of the world Athletics championship Budapest 2023 at National Athletics Centre on August 26, 2023 in Budapest, Hungary.



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CHAPTER



UNIT 5

The World Around Us

Famous Cities of the World

Try this quiz to brush up your knowledge of the world's cities.

1. You are in a large square. On one side is St. Basil's Cathedral. On another side is the GUM department store. Nearby is a walled fortress. In which city are you?



2. You are beside a large harbour. One of the world's largest steel span bridges is to your left. Behind you is a fantastic opera house with a roof of sails. In which city are you?

3. You are on the beach of a large city. Over looking this city are mountains, one of which is Sugarloaf Mountain. On another mountain is a large statue of Christ. In which city are you?



4. You are crossing the Grand Canal by the Rialto Bridge. You cannot see the traffic on the canal below as there are stores on each side of the bridge. If you could see this traffic it would include gondolas. In which city are you?

5. You are in Europe's highest capital. It is situated in the Pyrenees.



6. You are in the downtown area of a large city in North America. Rising high are the CN Tower and the two curved towers of City Hall. This city has an American League Baseball team and an NHL Hockey team. In which city are you?

7. You are looking at the remains of a church tower, all that is left of a church destroyed in World War II. A famous street in this city near the River Spree is called Unter den Linden. In which city are you?



8. You are looking at a statue of the Little Mermaid in the harbour of a city that has a large amusement park, Tivoli Gardens. In which city are you?

9. You are in a vast square. On one side of this square is the Forbidden City. Also located here is a mausoleum where a former leader of the country is on display. In which Asian capital are you?



10. You are standing on a bridge over the river Seine looking at Notre Dame Cathedral. Another church in this city is the Church of Sacre Coeur. In which city are you?



Adelaide is the capital city of South Australia and the fifth-largest city in Australia.



Activity

Find out about more cities of the world. Discuss in group.



CHAPTER 32 Economic Capital

Try this quiz to brush up your knowledge of the world of economic capital. Choose the correct option.

- I was the starting point of 2010 Tour de France. I was destroyed by German bombing during the Blitzkrieg, then rebuilt. I have served as the largest port city in Europe and the second in the world, in terms of service. Who am I?
 (a) Lisbon (b) Rotterdam
 (c) London (d) Marseillies
- I am the city closest to where the Prime Meridian and Equator intersect. I was originally a small fishing village. I am the chief port city of Ghana. Who am I?
 (a) Casablanca (b) Freetown
 (c) Tema (d) Accra
- I was the capital of Israel from 1948 until 1950. I house the majority of Israel's population in my metropolitan area. I was founded in 1909, making me one of the newest cities in what was to become Israel. Who am I?
 (a) Tel Aviv (b) Eilat
 (c) Ashqelon (d) Haifa
- I was the former capital of Cote D'Ivoire. I have been the chief manufacturing centre of Cote D'Ivoire. I reached a metropolitan population of over 5,000,000 in 2010. Who am I?
 (a) Abidjan (b) Lagos
 (c) Yaounde (d) Tangier



1



2



3



4



Activity

Prepare a presentation on five cities like these and present in the class.

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CHAPTER

Famous Town Squares in the World

A Town Square is an open public space commonly found in the heart of a traditional town used for community gatherings. Most town squares are suitable for open markets, music concerts, political rallies and other events that require firm ground.

Try this quiz to test your knowledge; answer the following.

1. The world's most famous square is located at the junction of Broadway and Seventh Avenue and stretching from West 42nd to West 47th streets. It's filled with video screens, LED signs, and flashing lights and is known to be a major tourists attraction. It is also known for its famous New Year's Eve ball drop.



2. It is located in the heart of Moscow. It is surrounded by Saint Basil's Cathedral, the State History Museum, Lenin's Mausoleum and the Kremlin. It is Russia's most recognizable landmark and is usually tourists first stop in Moscow.

3. It is the largest city square in the world. It is surrounded by Soviet-style monuments and government buildings. There is a ceremony for flag raising and lowering at dawn and dusk. It became famous due to the Square Massacre on June 4, 1989 in which troops with assault rifles and tanks inflicted thousands of casualties on unarmed civilians.



4. Located between Wenceslas Square and the Charles Bridge, it is often bursting with tourists and locals in the summer. Featuring various architectural styles includes the Gothic T'yn Cathedral and Baroque Saint Nicholas Church, the city square is an oasis for travellers tired of the narrow streets Prague.

The shell shaped Pizza del Campo is the principal public space of the historic centre of Siena, Italy. It is one of the Europe's greatest medieval city squares and is known worldwide for its beauty and architectural integrity.



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CHAPTER

A City by the Sea

Many of the world's cities sit on the edge of sea, and with this quiz it's up to you to determine the locations based on the clues. Choose the correct option.

- The Caribbean Sea is home to a number of capital cities including which of these on the island of Hispaniola?
 - (a) Belmopan
 - (b) Santo Domingo
 - (c) San Jose
 - (d) Managua
- Found at the tip of Sinai Peninsula in the Red Sea, What Egyptian city is sometimes considered part of Asia?
 - (a) Eilat
 - (b) Cairo
 - (c) Sharm el-Sheikh
 - (d) Aswan
- The largest city of the Caspian Sea, the world's largest land blocked sea (though may consider it a lake), is which of these world capitals known for its oil industry?
 - (a) Ankara
 - (b) Baku
 - (c) Almaty
 - (d) Tbilisi
- Which of these cities is NOT found on the Baltic Sea Coast?
 - (a) Stockholm
 - (b) Riga
 - (c) Saint Petersburg
 - (d) Rotterdam
- The westernmost part of the Mediterranean Sea is the Alborn Sea, which also includes the Strait of Gibraltar. Which European city can be found in this region?
 - (a) Algiers
 - (b) Malaga
 - (c) Marseille
 - (d) Palam de Mallorca

6. Which of these cities is found on the Sea of Japan?

- (a) Taipei
- (b) Vladivostok
- (c) Manila
- (d) Tokyo



7. Which of these Southern Hemisphere cities can be found on the edge of the Tasman Sea?

- (a) Darwin
- (b) Canberra
- (c) Sydney
- (d) Christchurch



8. The Black Sea is the major body of water for what Winter Olympic City?

- (a) Sochi
- (b) Cortina d' Ampezzo
- (c) Sarajevo
- (d) Innsbruck



9. Only one of these locations is in the South China Sea, but which is it?

- (a) Phnom Penh
- (b) Shanghai
- (c) Jakarta
- (d) Da Nang



10. Venice, Italy is on the Adriatic Sea, a smaller part of the Mediterranean. What walled city can also be found on this body of water?

- (a) Rome
- (b) Dubrovnik
- (c) Naples
- (d) Tirana



Activity

Find out about five Indian cities by the sea. Discuss in class.



Run Down the Line

There are several world cities which share the same longitude. Read the clues given the name of one of the cities and then answer the other city. Choose the correct option.

- Longitude 130° W : Vancouver in Canada is on the same longitude as this American city which lies on the Pacific Ocean. It has the largest community of Chinese people outside of China. Where am I?
(a) San Diego ☐ (b) Seattle ☐
(c) San Francisco ☐ (d) Los Angeles ☐
- Longitude 90° W : Havana in Cuba is on the same longitude as this American city which lies on the Mississippi River. Martin Luther King Jr was assassinated in this city outside room 306 at the Lorraine Motel. Where am I?
(a) Minneapolis ☐ (b) Kansas City ☐
(c) Nashville ☐ (d) Memphis ☐
- Longitude 80° W : Toronto in Canada is on the same longitude as this Colombian city which is home to two soccer stadiums, Estadio El Clampus and Estadio Metropolitano de Techo. Where am I?
(a) Quito ☐ (b) Punta Arenas ☐
(c) Bogota ☐ (d) Concepcion ☐
- Longitude 60° W : Montevideo in Uruguay is on the same longitude as this capital city of an Island in the Atlantic Ocean. This same island has been part of a tug of war battle between Argentina and Great Britain. Where am I?
(a) Goose Green ☐ (b) Port Howard ☐
(c) Darwin ☐ (d) Port Stanley ☐
- Longitude 10° W : Liverpool in England is on the same longitude as this African city which is home to Hassan II Mosque, one of largest mosques in the world. Where am I?
(a) Tunis ☐ (b) Marrakech ☐
(c) Casablanca ☐ (d) Rabat ☐

6. Longitude 10° E : also in Norway is on the same longitude as this city in the middle of the Mediterranean Sea. It was designated a UNESCO World Heritage Site in 1980 with history which dates back to the Knights of St. John of Jerusalem. Where am I?
- (a) Palermo ☐ (b) Cagliari ☐
(c) Palma ☐ (d) Valletta ☐
7. Longitude 40° E : Basra in Iraq is on the same longitude as this city on the Arabian Peninsula. The city is famous for King Fahd Road. Where am I?
- (a) Riyadh ☐ (b) Tehran ☐
(c) Kuwait City ☐ (d) Muscat ☐
8. Longitude 70° E : Mumbai in India is on the same longitude as this city in northern Pakistan. Alexander the Great and Genghis Khan used it as a route through to invade South Asia. Where am I?
- (a) Hyderabad ☐ (b) Chennai ☐
(c) Islamabad ☐ (d) Bangaluru ☐
9. Longitude 100° E : Phnom Penh in Cambodia is on the same longitude as this Mongolian city which has rail connections to Russia and China by way of the famous Trans-Siberian Railway. Where am I?
- (a) Kathmandu ☐ (b) Astana ☐
(c) Thimphu ☐ (d) Ulan Bator ☐
10. Longitude 100° E : Melbourne in Australia is on the same longitude as this city located on an island in the Pacific Ocean. One of the main languages is Tok Pisin, and the people who use this language call their city "Pot Mosbi". Where am I?
- (a) Port Talbot ☐ (b) Port Erin ☐
(c) Port-au-Prince ☐ (d) Port Moresby ☐



Activity

Find out about more cities like these. Discuss in class.



CHAPTER 36 National Symbols

Write the correct letters in the boxes to match the countries, with their national emblems and symbols.

- | | | |
|--------------------|---|---------------------------|
| 1. Egypt |  | (a) Iris, or fleur-de-lis |
| 2. England |  | (b) Tulip |
| 3. France |  | (c) Leek |
| 4. Ireland |  | (d) Lion |
| 5. Japan |  | (e) Lotus |
| 6. The Netherlands |  | (f) Chrysanthemum |
| 7. Scotland |  | (g) Maple Leaf |
| 8. Switzerland |  | (h) Bald Eagle |
| 9. Wales |  | (i) Edelweiss |
| 10. Canada |  | (j) Thistle |
| 11. Sri Lanka |  | (k) Rose |
| 12. The USA |  | (l) Shamrock |

India has a national calendar which is based on the Saka Era, with Chaitra as its first month.



Activity

Create an emblem for your class, what would it be? Make teams, discuss with the team members and create it. Display and discuss the emblems created by each group. Speak about why you choose the emblem.

Wonders of the World

In which country would you be if you visited the following wonders?

1 The Colosseum



2 Petra



3 Chichen Itza Versailles



4 Christ the Redeemer



5 Machu Picchu



6 The Great Wall of China



Helpbox



China, Peru, Brazil, Jordan, Italy, Mexico



St. Sophia is also called Hestia Sophia. Hestia Sophia means "holy wisdom" in Greek.



General Knowledge Quiz

1. What is the orange part of an egg called? _____
2. How many legs do insects have? _____
3. What is a baby Kangaroo called? _____
4. What is the closest planet to the Sun? _____
5. In which country can you find the Eiffel tower. _____
6. How many days are there in a year? _____
7. How many players are in a soccer team? _____
8. Where do polar bears live? _____
9. Which is faster, light or sound? _____
10. How many letters are in the English alphabet? _____
11. What is the name of a shape with 5 sides? _____
12. How many continents are there? _____
13. What is the name of the tallest mountain on earth? _____
14. The planet Mars is also known as what? The planet. _____
15. How many months have 31 days? _____
16. How many colours of the rainbow are there? _____
17. My mother's mother is my what? _____
18. Which animal lays the largest eggs? _____
19. How many zeros are in ten thousand? _____
20. In which country would you find the Sydney opera House? _____



The Wonderful World of Mythology

The wonderful world of Mythology. What do you know about it? Take this quiz.

1. Who was on a quest to find the Golden Fleece?
2. Who is gloomy king of the underworld that was so enchanted by Orpheus music?
3. Who tamed Pegasus?
4. In Hindu mythology, who is the God of sky, water and the oceans?
5. Which mythical country in South America was believed to be rich in gold?
6. According to Japanese Shinto religion, who is Amaterasu?
7. According to Greek myths, who stole fire from the heavens and brought it to Earth?
8. In the epic Mahabharata, the ornaments Kavacha and Kundala were given to whom by the Sun God?
9. Which God designed the Rama's mighty bow?

Helpbox

Varuna, Sun Goddess, Hades, Visvakarma, El Dorado, Jason, Prometheus, Bellerophon, Karna



Some hunters in the North American forests have reported seeing a huge hairy ape-like figure whom they have named Sasquatch or Bigfoot.

**Activity**

Find 10 more myths around the world. Discuss in class.



Borrowed Words

Complete the following by writing the name of languages from which the given words have originated, and the meaning of words.

WORD	LANGUAGE	MEANING
1. Abandon	_____	_____
2. Agile	_____	_____
3. Embarrassment	_____	_____
4. Lunar	_____	_____
5. Nocturnal	_____	_____
6. Thug	_____	_____
7. Bonsai	_____	_____
8. Dinghy	_____	_____
9. Wanderlust	_____	_____
10. Gung-ho	_____	_____
11. Mammoth	_____	_____
12. Chauffeur	_____	_____



Schadenfreude is a German word which means to take delight in the misfortunes of others.



Activity

Find out twenty more words like these, remember them and discuss in class.

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CHAPTER

Women Writers

Match these women writers with the books they have written by writing the correct letters in the boxes.

1. Anita Desai



2. Anita Nair



3. Pearl S Buck



4. Anne Bronte



5. Jhumpa Lahiri



6. Kiran Desai



7. Jane Austen



8. Mahasweta Devi



9. Amrita Pritam



10. Louisa May Alcott



11. Agatha Christie



12. Anna Sewell



(a) The Good Earth, Dragon

(b) Inheritance of Loss

(c) Angles Grey

(d) Northanger Abbey

(e) The Village by the Sea

(f) Little women, Little men

(g) Black Beauty

(h) The Puffin Book of
Magical Indian Myths

(i) Rudaoli

(j) The Namesake

(k) Kagaz Te Canvas

(l) Death on the Nile



Activity

Find the names of some Indian poetess. Read out their poems in class.



Books and Authors

Choose the correct options.

1. Who wrote the famous book– ‘We the people’?
(a) T.N. Kaul (b) J.R.D. Tata
(c) Khuswant Sinha (d) Nani Palkhivala
2. Who is the author of the book ‘Nineteen Eighty Four’?
(a) Thomas Hardy (b) Emile Zola
(c) George Orwell (d) Walter Scott
3. Which is not written by Munshi Premchand?
(a) Gaban (b) Godan
(c) Guide (d) Manasarovar
4. Who is the author of the book ‘Forbidden Verses’?
(a) Salman Rushdie (b) Abu Nuwas
(c) Ms. Taslima Nasrin (d) D.H. Lawrence
5. Who wrote the line : ‘A thing of beauty is a joy forever’?
(a) John Keats (b) Robert Browning
(c) P.B. Shelley (d) William Wordsworth
6. Which of the book was written by Graham Greene?
(a) O’ Jerusalem (b) The Power and Glory
(c) Crisis in India (d) Of Human Bondage
7. The famous book ‘Anand Math’ was authored by :
(a) Sarojini Naidu (b) Bankim Chandra Chattopadhyay
(c) Sri Aurobindo (d) Rabindranath Tagore



Activity

Read 5 books of Indian authors, discuss it in class, explaining what you liked about the book and what you didn’t.



‘War and Peace’ was written by Leo Tolstoy.



Autobiographies

A history of a person's life written or told by that is called the autobiography.

Match the following autobiographies with their writers by writing the correct letters in the boxes.

- | | | |
|------------------------------------|---|-----------------------|
| 1. The Story of My Life |  | (a) Buzz Aldrin |
| 2. The Diary of a young Girl |  | (b) Mahatma Gandhi |
| 3. Reaching for the Moon |  | (c) Helen Keller |
| 4. Moments |  | (d) Cristiano Ronaldo |
| 5. My Experiments with Truth |  | (e) Anne Frank |
| 6. My life— A Spoken Autobiography |  | (f) RK Narayan |
| 7. Sunny Days |  | (g) Benazir Bhutto |
| 8. Mein Kampf |  | (h) Fidel Castro |
| 9. My Days |  | (i) Adolf Hitler |
| 10. Daughter of the East |  | (j) Sunil Gavaskar |
| 11. Dreams of My father |  | (k) Charlie Chaplin |
| 12. My Autobiography |  | (l) Barack Obama |

- David Copperfield by Charles Dickens and Catcher in Rye by JD Salinger are called fictional autobiographies.
- Recently Sachin Tendulkar has also launched his autobiography "Playing it my Way."



Activity

Read out 10 autobiographies of some Indians and discuss in class.



Nom De Plume

A pen name or nom de plume is a pseudonym adopted by a author. A pen name may be used to make the author's name more distinctive.

Some writers use a different name to write. Match the pen name of the writers with their real name.

PEN NAME

1. Chanakya
2. Sue Denim
3. Mary West Maccott
4. Mark Twin
5. Saki
6. Herge
7. O Henry
8. Bhanusingha
9. George Eliot
10. Silence Dogood
11. Boz
12. Gulzar



REAL NAME

- (a) Dar Pilkey
- (b) Jawaharlal Nehru
- (c) William Sydney Porter
- (d) Charles Dickens
- (e) Sampooran Singh Kalra
- (f) Samuel Langhorne Clemens
- (g) Agatha Christie
- (h) Georges Remi
- (i) Rabindranath Tagore
- (j) Hector Hugh Munro
- (k) Benjamin Franklin
- (l) Mary Ann Evans



Activity

Find out more nom de plume and discuss in class.



Travelogue

Travelogue is a documentary film, television program or a form of travel documentary that describe travel in general.

If you like to travel and see new places, you may also enjoy reading travelogues. Name the following with their authors by writing the correct answers with the help of help box.

1. A travelogue about the author's out-of the way destinations to far-flung islands in North Korea, Argentina, Cuba, Iceland, Bhutan, Vietnam, Paraguay, and Australia.

2. An account of the author's 7- year journey through the Islamic world during the 11th century.

3. A travelogue about the historical capital of India, Delhi, written by the author during his stay in New Delhi.

4. A historical account of the experiences of a European in the empire of the Mongol leader Kublai Khan.

5. A humorous account of the author's travels through all corners of Great Britain.

Helpbox



City of Djinns by William Dalrymple, IL Milione by Marco Polo, Notes from a Small Island: Travels in the UK by Bill Bryson, Safarnama by Nasir Khusraw, Falling off the Map: Some Lonely Places of the World by Pico Iyer

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CHAPTER

The Art of Beautiful Writing

Take this quiz, answer the following by choosing the correct option.

- What type of paper is easiest to use to practice the spacing between your letters, words and lines?
 (a) Graph Paper (b) Parchment
 (c) Tissue Paper (d) Tracing Paper
- What is the name of the miniature sketch used to work out your design before putting pen to paper?
 (a) Beginner's (b) Primary
 (c) Thumbnail (d) Preparatory
- What pens have separate interchangeable nibs that you need to use with an inkpot?
 (a) Dipping pens (b) Roller ball
 (c) Potted pens (d) Quill pens
- What is the layout called that requires equal left and right margins?
 (a) Full (b) Equal
 (c) Justified (d) Rigid
- What border is usually slightly broader than the others?
 (a) Left (b) Bottom
 (c) Right (d) Top
- What script can you not learn to write in calligraphy?
 (a) Italic majuscules (b) Copperplate
 (c) Lombardic (d) Topaz
- What type of pen is more suitable to temporary projects?
 (a) Ball point (b) Quill pen
 (c) Gold-plated (d) Felt tip
- What type of ink is most suitable if you are going to paint over the letters?
 (a) Non-water proof ink (b) Walnut ink
 (c) Gouache (d) Water proof ink

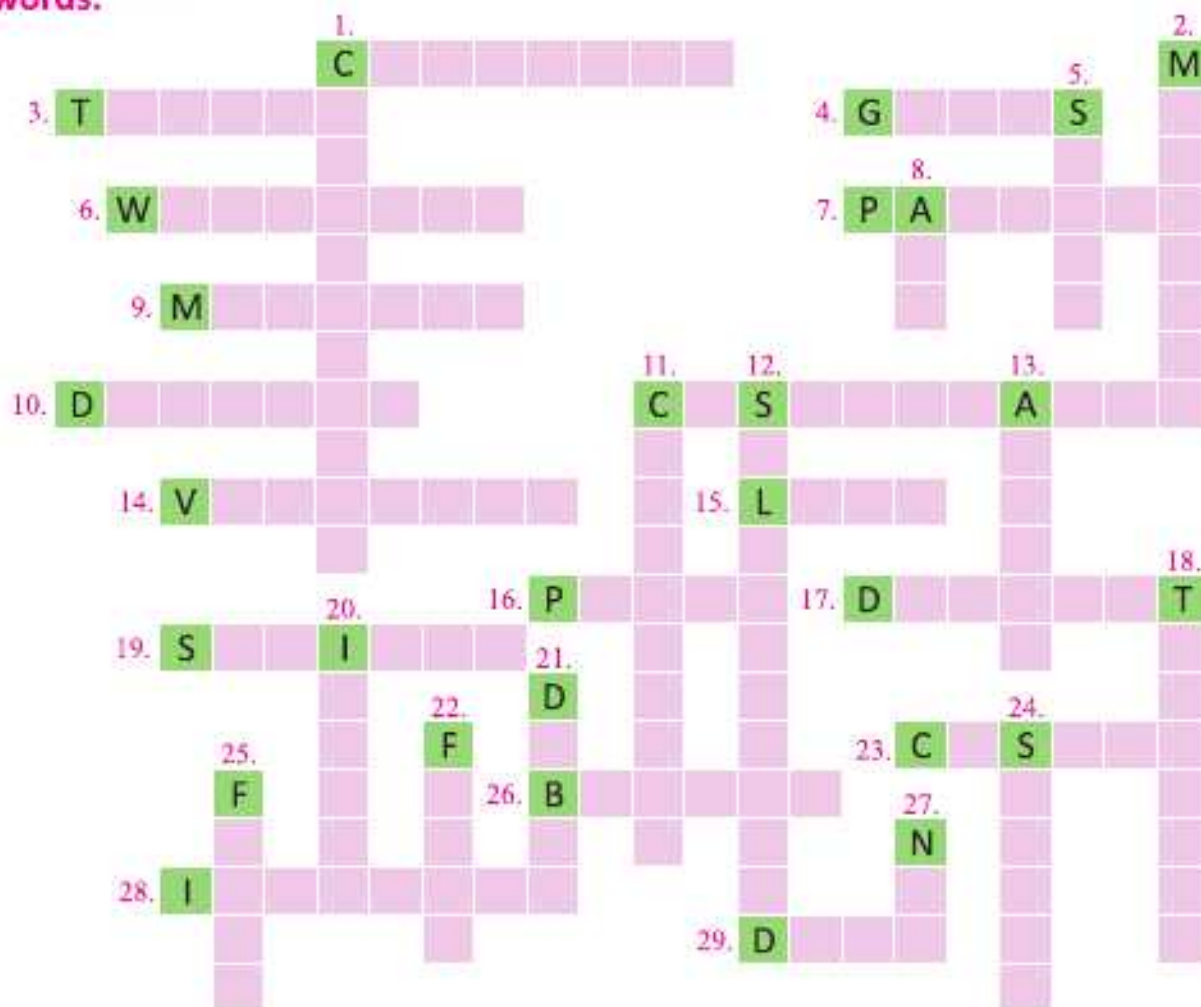


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CHAPTER

Banking and Financial Terms

How well do you know banking and financial terms? Complete this crossword to find out. The numbers given alongside each clue below will tell you how many letters there are in the words.



Across :

1. A bank account used to pay everyday expenses (8)
3. A bank employee (6)
4. Total income before tax is deducted (5)
6. Take money out of the bank (8)
7. Money that is paid back at regular intervals after it is borrowed monthly (7)



9. Every month (7)
10. To put money in the bank (7)
11. Money borrowed using a credit card (4, 7)
14. An interest rate that changes with time (8)
15. Money that has been borrowed (4)
16. A bank's main interest rate (5)
17. Not pay a loan (7)
19. A bank account that is used to save money (7)
23. Guarantee a loan for somebody else (6)
26. The opposite of lend (6)
28. A monthly fee on borrowed money (8)
29. If money is borrowed, it results in this (4)



Down :

1. The maximum one can borrow on a credit card (6, 5)
2. A house loan (8)
5. The opposite of save (5)
8. A cash machine (3)
11. VISA, Master Card, etc. are examples of this (6, 4)
12. When one works for oneself, and is not employed anywhere (4, 8)
13. Yearly (6)
18. To move money from one bank account to another (8)
20. Money that is earned (6)
21. Money that is withdrawn from an account (5)
22. An interest rate that stays the same always (5)
24. What one earns each month at a job (6)
25. The money that one has (5)
27. Income remaining after the taxes are paid (3)



Activity

Find out more terms and discuss it in class.



Alpha-Numeric Series

Find the next term in the alpha-numeric series.

1. Z1A, X2D, V6G, T21J, R88M, P445P,?

(a) N2676S

(c) T2670N

(b) N2676T

(d) T2676N

2. D-4, F-6, H-8, J-10, ____?

(a) K-12, M-13

(c) L-12, N-14

(b) L-12, M-14

(d) K-12, M-14

3. 2B, 4C, 8E, 14H,?

(a) 16K

(c) 20L

(b) 20I

(d) 22L

4. 3F, 6G, 11I, 18L,?

(a) 21O

(c) 25P

(b) 25N

(d) 27P

5. W-144, U-121, S-100, Q-81,?

(a) U-122

(c) O-64

(b) V-121

(d) V-128

6. 2, A, 9, B, 6, C, 13, D,?

(a) 9

(c) 12

(b) 10

(d) 19

7. KM5, IP8, GS11, EV14,?

(a) BX17

(c) CY17

(b) BY17

(d) CY18

8. C4X, F9U, I16R,?

(a) K25P

(c) L25O

(b) L25P

(d) L27P

9. 2Z5, 7Y7, 14X9, 23W11, 34V13,?

(a) 27U24

(c) 47U15

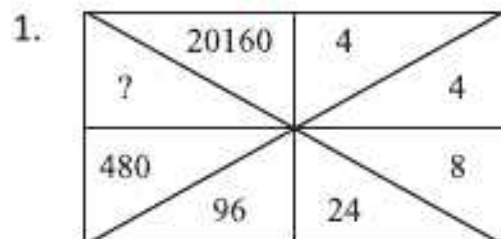
(b) 45U15

(d) 47V14



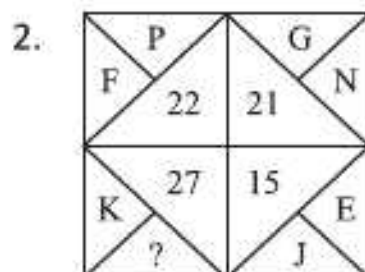
Inserting the Missing Character

Find the missing character from among the given alternatives.



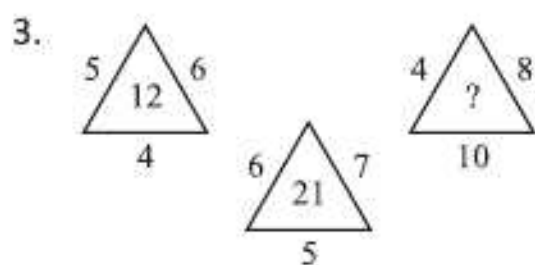
- (a) 860
(c) 2880

- (b) 1140
(d) 3240



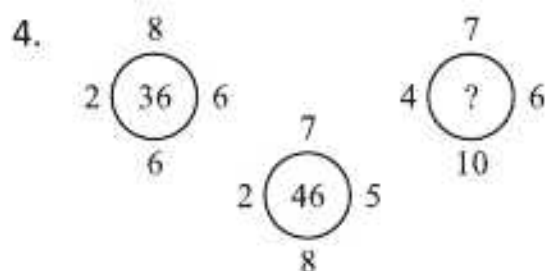
- (a) M
(c) Q

- (b) P
(d) S



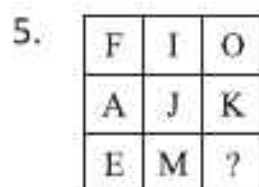
- (a) 22
(c) 32

- (b) 30
(d) 40



- (a) 42
(c) 48

- (b) 46
(d) 50



- (a) P
(c) S

- (b) R
(d) V



CHAPTER 50 Alphabetical Order of Words



In each of the following questions, arrange the given words in the sequence in which they occur in the dictionary and then choose the correct sequence.

1. 1. Brook, 2. Bandit, 3. Boisterous, 4. Baffle, 5. Bright
(a) 2, 4, 3, 1, 5 (b) 2, 4, 3, 5, 1 (c) 4, 2, 3, 1, 5 (d) 4, 2, 3, 5, 1
2. 1. Wound, 2. Writer, 3. Whiter, 4. Worst, 5. Worked
(a) 1, 4, 3, 5, 2 (b) 2, 1, 3, 4, 5 (c) 3, 5, 4, 1, 2 (d) 5, 3, 2, 1, 4
3. 1. Preach, 2. Praise, 3. Precinct, 4. Precept, 5. Precede
(a) 2, 1, 5, 4, 3 (b) 2, 1, 3, 4, 5 (c) 1, 2, 5, 4, 3 (d) 2, 5, 1, 4, 3
4. 1. Select, 2. Seldom, 3. Send, 4. Selfish, 5. Seller
(a) 1, 2, 4, 5, 3 (b) 2, 1, 5, 4, 3 (c) 2, 1, 4, 5, 3 (d) 2, 5, 4, 1, 3
5. 1. Eagle, 2. Earth, 3. Eager, 4. Early, 5. Each
(a) 1, 5, 2, 4, 3 (b) 2, 1, 4, 3, 5 (c) 2, 3, 5, 4, 1 (d) 5, 3, 1, 4, 2
6. 1. Page, 2. Pagan, 3. Palisade, 4. Pageant, 5. Palate
(a) 1, 4, 2, 3, 5 (b) 2, 4, 1, 3, 5 (c) 1, 4, 2, 5, 3 (d) 2, 1, 4, 5, 3
7. 1. Wrinkle, 2. Wriggle, 3. Writhe, 4. Wretch, 5. Wrath
(a) 4, 5, 1, 2, 3 (b) 5, 4, 2, 1, 3 (c) 4, 2, 5, 1, 3 (d) 5, 2, 1, 3, 4
8. 1. Repoint, 2. Reptile, 3. Repent, 4. Repute, 5. Report
(a) 3, 1, 5, 2, 4 (b) 3, 5, 1, 2, 4 (c) 5, 1, 4, 3, 2 (d) 5, 4, 1, 3, 2
9. 1. Spruce, 2. Spume, 3. Spree, 4. Spurt, 5. Sprawl
(a) 5, 3, 1, 2, 4 (b) 1, 2, 3, 4, 5 (c) 3, 5, 1, 4, 2 (d) 5, 4, 3, 2, 1
10. 1. Protein, 2. Problem, 3. Proverb, 4. Property, 5. Project
(a) 1, 2, 3, 4, 5 (b) 2, 1, 4, 3, 5 (c) 2, 5, 4, 1, 3 (d) 3, 4, 5, 2, 1



Situation Reaction Test

Choose the best response which shall present you as a good person.

1. Your maid has invited you to her daughter's wedding. You would :
 - (a) Completely ignore her
 - (b) Attend the wedding
 - (c) Buy a gift for her daughter
 - (d) Congratulate her and make up some excuse for not being able to attend
2. Your friends like smoking and influence you to do the same. You will :
 - (a) Smoke only because your friends are smoking
 - (b) Refuse to smoke
 - (c) Smoke but only in their presence
 - (d) Refuse and lie to them that you have asthma
3. When you see a blind man trying to cross the road, you :
 - (a) Ask someone to help him
 - (b) Go and help him
 - (c) Wait till he crosses the road
 - (d) Ignore and move on
4. "No risk no gain". You :
 - (a) Feel that risk means no gain
 - (b) Believe that this slogan is correct
 - (c) Feel it is foolish to accept unnecessary risk
 - (d) Feel that risk may be taken only after judging the situation thoroughly
5. In a bus you realise that someone has left his/her wallet. You would :
 - (a) Give the money to the beggar
 - (b) Hand it over to the bus conductor
 - (c) Try to find his/her contact number and inform the owner
 - (d) Leave the wallet as it is



Choosing the Odd Word

Choose the word which is least like the other words in the group.

1. (a) Zebra (b) Lion (c) Tiger (d) Horse (e) Giraffe
2. (a) Parrot (b) Bat (c) Crow (d) Sparrow (e) Pigeon
3. (a) Copper (b) Zinc (c) Brass (d) Aluminium (e) Iron
4. (a) Apple (b) Marigold (c) Rose (d) Lily (e) Lotus
5. (a) January (b) May (c) July (d) August (e) November
6. (a) Amethyst (b) Ruby (c) Marble (d) Sapphire (e) Diamond
7. (a) Ginger (b) Onion (c) Beetroot (d) Coriander (e) Potato
8. (a) Bake (b) Peel (c) Fry (d) Boil (e) Roast
9. (a) Pistol (b) Sword (c) Gun (d) Rifle (e) Cannon
10. (a) Cathedral (b) Mosque (c) Church (d) Monastery (e) Temple
11. (a) Pear (b) Apple (c) Litchi (d) Guava (e) Orange
12. (a) Curd (b) Butter (c) Oil (d) Cheese (e) Cream
13. (a) Assassinate (b) Kill (c) Kidnap (d) Stab (e) Murder
14. (a) Shehnai (b) Bagpipe (c) Flute (d) Sitar (e) Harmonica
15. (a) Football (b) Volleyball (c) Cricket (d) Chess (e) Hockey



Unit-1 : The Natural World

Chapter-1

1. Dal Lake, 2. Chilika Lake, 3. Mansarovar Lake, 4. Hussain Sagar Lake, 5. Lonar Lake, 6. Loktak Lake, 7. Lake Baikal.

Chapter-2

1. Skull, 2. Milk, 3. Red Kangaroo, 4. Hibernation, 5. Birds, 6. Otter, 7. Walrus, 8. Scavengers, 9. Marsupials.

Chapter-3

1. 10 million years, 2. 5500, 3. 1300, 4. under the ocean, 5. 1150-1200 degrees Celsius, 6. A circle of volcanoes that surrounds the Pacific Ocean, 7. Navada del Ruiz, 8. Mount Tambora (Indonesia), 9. Mauna Loa; Hawaii, 10. Mount Vesuvius.

Chapter-4

1. a, 2. b, 3. b, 4. a, 5. b, 6. b, 7. b, 8. a.

Chapter-5

1. The Blue Dragon, 2. The Red-Lipped Batfish, 3. The Bush Viper, 4. The Blue Parrot Fish, 5. The Indian Purple Frog.

Chapter-6

1. Renewable Energy, 2. Fuel Cell, 3. Bicycle, 4. Natural Gas, 5. Earth Day, 6. Organic, 7. Top Soil, 8. Ozone Layer, 9. Reduce; Reuse; Recycle, 10. Kyoto Treaty, 11. Green House Gas, 12. Hydrogen, 13. Sustainable, 14. Solar Energy.

Unit-2 : The World of Science

Chapter-8

1. d, 2. j, 3. i, 4. g, 5. a, 6. h, 7. e, 8. c, 9. b, 10. f.

Chapter-9

1. platelets, 2. gall bladder, 3. keratin, 4. cardiology, 5. transmit neural, 6. skin, 7. femur, 8. posterior, 9. Cavities.

Chapter-10

1. Merck & Co, 2. Computer Mouse, 3. Abio Med, 4. Adidas, 5. US Air Force.

Chapter-11

1. George Lucas, 2. Issac Assimov, 3. The Wacknowski Brothers, 4. S. Shanker.

Chapter-12

- A:** 1. T, 2. T, 3. T, 4. F, 5. F, 6. T

- B:** 1. Finger prints, 2. Dolly, 3. Double helix, 4. Heredity, 5. Chromosomes, 6. Identical, 7. DNA, 8. Blood, 9. Tissues.

Chapter-13

1. b, 2. a, 3. a, 4. b, 5. c, 6. a, 7. a, 8. a, 9. c, 10. c, 11. b, 12. b, 13. c, 14. c, 15. b, 16. b.

Chapter-14

1. electrons, 2. theory of relativity, 3. Nuclear fission, 4. quarks, 5. nucleus, 6. binding energy.

Chapter-15

1. b, 2. b, 3. c, 4. c, 5. b, 6. a, 7. c, 8. b, 9. a.

Chapter-16

1. Spanish Influenza, 2. Peru, 3. River Kosi.

Unit-3 : Our Country : India

Chapter-17

1. (c), 2. (d), 3. (b), 4. (a), 5. (d), 6. (b), 7. (d), 8. (d), 9. (b), 10. (c), 11. (c), 12. (a)

Chapter-18

1. (b), 2. (b), 3. (i), 4. (a), 5. (a), 6. (b), 7. (c), 8. (a), 9. (b), 10. (a)

Unit-4 : Sports and Entertainment

Chapter-20

1. Deepa Mehta, 2. Yash Chopra, 3. Guru Dutt, 4. Raj Kapoor, 5. Mira Nair.

Chapter-21

1. c, 2. a, 3. c, 4. c, 5. c, 6. b, 7. c.

Chapter-22

1. Ravi Shankar, 2. Mozart, 3. Hariprasad Chaurasia, 4. Georg Friedrich Handel, 5. Pyotr Tachikovsky.

Chapter-23

1. a, 2. b, 3. c, 4. b, 5. c, 6. a, 7. d, 8. b, 9. a, 10. c, 11. a, 12. d, 13. b, 14. c.

Chapter-24

1. b, 2. a, 3. c, 4. d, 5. b, 6. a, 7. c, 8. d, 9. a, 10. b, 11. a, 12. d, 13. b.

Chapter-25

1. Swimming : Arjuna Award, 2. Badminton : Arjuna Award, 3. Swimming : Padma Shri Award, 4. Archery : Arjuna Award, 5. Cricket : Arjuna Award, 6. Women's Boxing : Rajiv Gandhi Khel Ratna, 7. Track and Field : Arjuna Award, 8. Badminton : Rajiv Gandhi Khel Ratna, 9. Tennis : Padma Shri Award, 10. Chess : Arjuna Award.

Chapter-26

1. Kavita Chahal, 2. Virat Kohli, 3. Dharmender Dalal, 4. Poornima Mahato, 5. Raj Singh, 6. Narinder Singh Saini, 7. Mary D'souza, 8. Syed Ali, 9. Anil Mann, 10. Ronjan Sodhi.

Chapter-27

1. Barcelona : Spain, 2. India, 3. Kuala Lumpur : India, 4. Pakistan : Holland, 5. Mumbai : Fifth, 6. London, 7. Holland : Pakistan, 8. Holland : Fifth, 9. Holland : Ninth, 10. Australia : Tenth, 11. Germany, 12. Australia, 13. Australia.

Chapter-28

1. d, 2. b, 3. c, 4. d, 5. a, 6. c, 7. d, 8. b.

Chapter-29

1. b, 2. a, 3. a, 4. c, 5. d, 6. b, 7. a, 8. b.

Unit-5 : The World Around Us

Chapter-31

1. Moscow, 2. Sydney, 3. Rio de Janeiro, 4. Venice, 5. Andorra la Vella, 6. Toronto, 7. Berlin,

8. Copenhagen, 9. Beijing, 10. Paris.

Chapter-32

1. b, 2. c, 3. a, 4. a.

Chapter-33

1. Times Square; New York; United States, 2. Red Square; Moscow; Russia, 3. Tiananmen Square; Beijing; China, 4. Old Town Square; Prague; Czech Republic.

Chapter-34

1. b, 2. c, 3. b, 4. d, 5. b, 6. b, 7. c, 8. a, 9. d, 10. b.

Chapter-35

1. c, 2. d, 3. c, 4. d, 5. c, 6. d, 7. a, 8. c, 9. d, 10. d.

Chapter-36

1. e, 2. k, 3. a, 4. e, 5. f, 6. b, 7. j, 8. i, 9. c, 10. g, 11. d, 12. h.

Chapter-37

1. Italy, 2. Jordan, 3. Mexico, 4. Brazil, 5. Peru, 6. China

Chapter-38

1. Yolk, 2. 6, 3. joey, 4. Mercury, 5. France, 6. 365, 7. 9, 8. Arctic regions, 9. Light waves, 10. 26, 11. Pentagon, 12. Seven, 13. Mount Everest, 14. Red planet, 15. 7, 16. 7, 17. Self, 18. Ostrich, 19. Four, 20. Australia.

Unit-6 : Language and Literature

Chapter-39

1. Jason, 2. Hades, 3. Bellerophon, 4. Varuna, 5. El Dorado, 6. Sun Goddess, 7. Prometheus, 8. Karna, 9. Visvakarma.

Chapter-40

1. French : To give up control, 2. French : Having the ability to move quickly, 3. French : The state of discomfort, 4. Latin : To the moon, 5. Latin : To a creature, 6. Hindi : Criminal, 7. Japanese : A small tree, 8. Sanskrit : A small rowing boat, 9. German : A yearning to travel, 10. Chinese : To be overly eager, 11. Russian : Large, 12. French : Driver.

Chapter-41

1. e, 2. h, 3. a, 4. c, 5. j, 6. b, 7. d, 8. i, 9. k, 10. f, 11. l, 12. g.

Chapter-42

1. d, 2. c, 3. c, 4. b, 5. a, 6. b, 7. b.

Chapter-43

1. c, 2. e, 3. a, 4. d, 5. b, 6. h, 7. j, 8. i, 9. f, 10. g, 11. l, 12. k.

Chapter-44

1. b, 2. a, 3. g, 4. f, 5. j, 6. h, 7. c, 8. i, 9. l, 10. k, 11. d, 12. e.

Chapter-45

1. Falling off the Map : Some Lonely Places of the World by Pico Iyer, 2. Safarnama by Nasir Khusraw, 3. City of Djinns by William Dalrymple, 4. IL Milione by Marco Polo, 5. Notes from a Small Island : Travels in the UK by Bill Bryson.

Chapter-46

1. a, 2. c, 3. a, 4. c, 5. b, 6. d, 7. d, 8. d.

Chapter-47

Across : 1. CHECKING, 3. TELLER, 4. GROSS, 6. WITHDRAW, 7. PAYMENT, 9. MONTHLY, 10. DEPOSIT,

11. CASH ADVANCE, 14. VARIABLE, 15. LOAN, 16. PRIME, 17. DEFAULT, 19. SAVINGS, 23. COSIGN, 26. BORROW, 28. INTEREST, 29. DEBT.

Down : 1. CREDIT LIMIT, 2. MORTGAGE, 5. SPEND, 8. ATM, 11. CREDIT CARD, 12. SELF EMPLOYED, 13. ANNUAL, 18. TRANSFER, 20. INCOME, 21. DEBIT, 22. FIXED, 24. SALARY, 25. FUNDS, 27. NET.

Unit-7 : Brain Teaser

Chapter-48

1. a, 2. c, 3. d, 4. d, 5. c, 6. b, 7. c, 8. c, 9. c.

Chapter-49

1. c, 2. b, 3. c, 4. b, 5. b.

Chapter-50

1. d, 2. c, 3. a, 4. c, 5. d, 6. d, 7. b, 8. a, 9. a, 10. c.

Chapter-51

1. c, 2. b, 3. b, 4. d, 5. c.

Chapter-52

1. d, 2. b, 3. c, 4. a, 5. e, 6. c, 7. d, 8. b, 9. b, 10. d, 11. e, 12. c, 13. c, 14. d, 15. d.